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Family Name ....

Given Name ....

Student No. ....

Signature ....

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Term 1, 2019

ELEC1111
Electrical and Telecommunications Engineering

TIME ALLOWED:	**2 hours**
TOTAL MARKS:	**100**
TOTAL NUMBER OF QUESTIONS:	**5**

THIS EXAM CONTRIBUTES 50% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 6 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

a. **(10 marks)** For the circuit shown in Figure 1,

   i. **(7 marks)** Calculate nodal voltages v_a, v_b and v_c.

   ii. **(3 marks)** Find the power supplied/absorbed by the dependent voltage source.

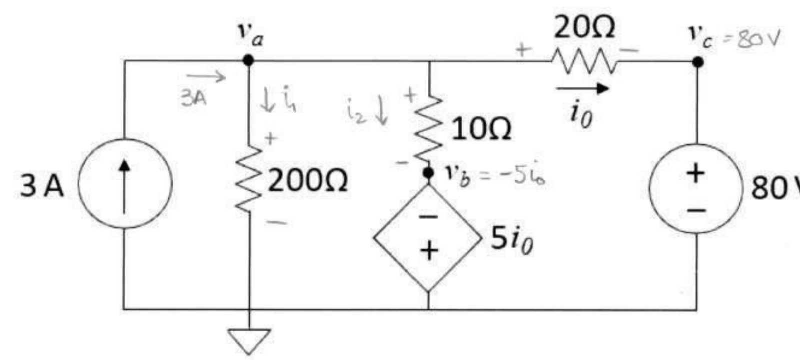


Figure 1

i)

KCL @ node a : $3 - i_1 - i_2 - i_0 = 0$

$$3 - \frac{V_a - 0}{200} - \frac{V_a - (-5i_0)}{10} - i_0 = 0 \xrightarrow[i_0 = \frac{V_a - 80}{20}]{} 3 - \frac{V_a}{200} - \frac{V_a}{10} - \frac{5}{10}\left(\frac{V_a - 80}{20}\right) - \frac{V_a - 80}{20} = 0$$

Simplify and solve for V_a:

$$3 - \left(\frac{1}{200} + \frac{1}{10} + \frac{5}{200} + \frac{1}{20}\right)V_a + \frac{80\times 5}{200} + \frac{80}{20} = 0$$

$$-\frac{1+20+5+10}{200}V_a + \frac{600+400+800}{200} = 0 \;;\; -36V_a + 1800 = 0 \Rightarrow \boxed{V_a = \frac{-1800}{-36} = 50\text{ V}}$$

Also, from the circuit:

$\boxed{V_c = 80V}$

$$V_b = -5i_0 = -5\,\frac{V_a - 80}{20} \quad -5\times \overset{= i_0 = -1.5A}{\frac{50-80}{20}} = 7.5\text{ V} \Rightarrow \boxed{V_b = 7.5V}$$

ii) $\boxed{P_{ds} = -V_{ds}\times i_{ds}} = -5i_0 \times i_2 = -5(-1.5)\times 4.25 = 7.5\times 4.25 = \boxed{31.875\text{ W}}$ (source is absorbing power)

$i_0 = -1.5A$

$$i_2 = \frac{V_a - (-5i_0)}{10} = \frac{50 + 5\times(-1.5)}{10} = \frac{50 - 7.5}{10} = 4.25A$$

b. **(10 marks)** The circuit shown in Figure 2 is being used to power a hot water heater. The heater can be modelled as a single resistor.

i. **(4 marks)** Calculate the Thevenin voltage at terminals *a-b*.

ii. **(4 marks)** Calculate the heater resistance that will ensure the maximum transfer of power from the circuit to the heater.

iii. **(2 marks)** Find the maximum power that can be delivered to the heater.

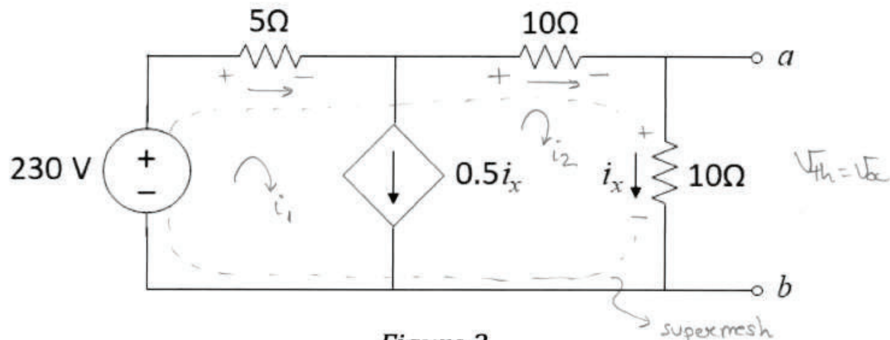


Figure 2

i)

$V_{th} = V_{oc} = 10\,i_x$

Using mesh analysis:

$-230 + 5i_1 + 10i_2 + 10i_2 = 0$ $\Rightarrow$ $5i_1 + 20i_2 = 230$ $\Rightarrow$ $5i_1 + 20i_2 = 230$ ①

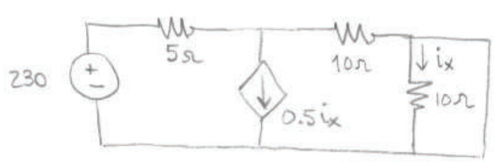
$i_1 - i_2 = 0.5i_x$ where $i_x = i_2$ $\Rightarrow$ $i_1 - i_2 - 0.5i_2 = 0$ $\Rightarrow$ $i_1 - 1.5i_2 = 0$ ② $(\times -5)$

$$\begin{array}{r} \cancel{5i_1} + 20i_2 = 230 \\ -\cancel{5i_1} + 7.5i_2 = 0 \\ \hline 27.5i_2 = 230 \end{array}$$

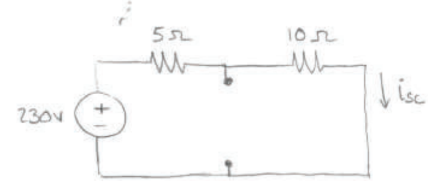
$\Rightarrow i_2 = i_x = 8.363\ A \Rightarrow \boxed{V_{th} = 10i_x = 83.63V}$

ii)

$R_{heater} = R_{Th} = V_{th}/i_{sc}$



$\longrightarrow$ $i_x = 0$ (10Ω resistor is bypassed)



$i_{sc} = 230/15 = 15.333\,A$

$\boxed{R_{th} = V_{th}/i_{sc} = \frac{83.63\ V}{15.333\ A} = 5.45\ \Omega = R_{heater}}$

iii)

$\boxed{P_{max} = \frac{V_{th}^2}{4 \times R_{th}} = \frac{83.63^2}{4 \times 5.45} = 320.82\ W}$

QUESTION 2 [20 marks]

a. **(8 marks)** Find voltage v, current i, and the energy stored in the capacitor in the circuit of Figure 3 under DC conditions.

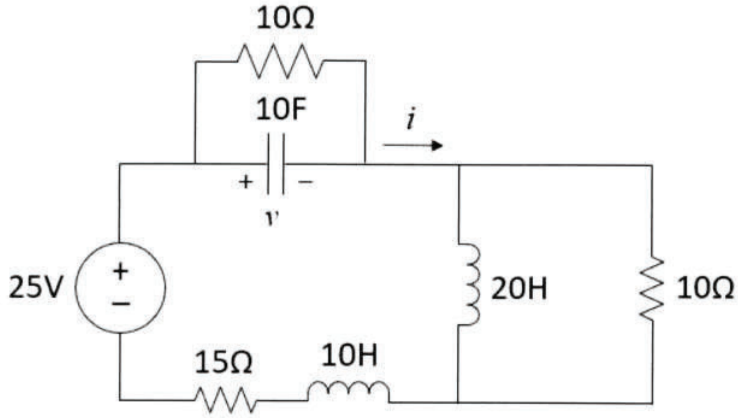
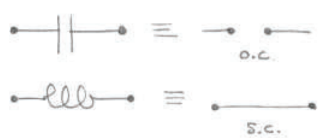


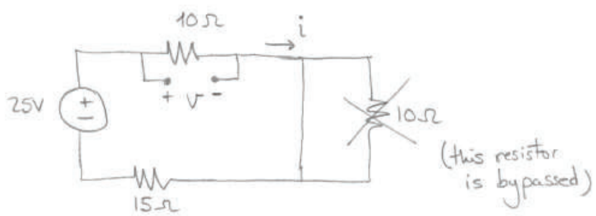
Figure 3

DC conditions :



(open circuit)

(short circuit)



$\boxed{i = 25/25 = 1\,A}$

$\boxed{v = 10i = 10\,V}$

$\boxed{w_c = \frac{1}{2}Cv^2 = 0.5 \times 10 \times 10^2 = 500\,J}$

b. (12 marks) The circuit below is used to take signals from a Hi-Fi system and amplify them so that they can drive a subwoofer. When first turned on, the amplifier fails and applies +15V DC directly to the output (v_a). After 30 milliseconds, the speaker protection circuit operates Sw1 to disconnect the speaker and connect it to ground (i.e., Sw1 changes from position '1' to position '2').

Taking the time the amplifier was turned on as t=0 seconds, calculate the voltage across the capacitor, v_C, for the first 60 milliseconds (i.e. for $0 < t \le 30$ ms and $30\text{ ms} < t \le 60$ ms).

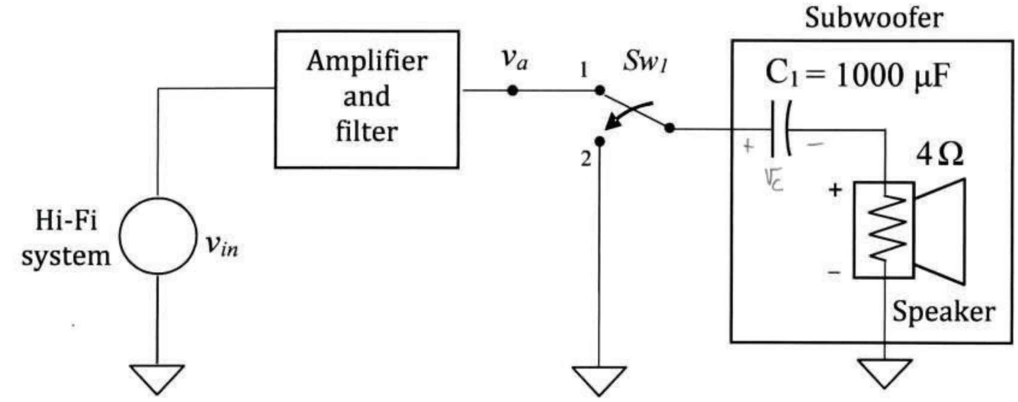
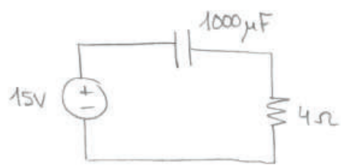


Figure 4

- $0 < t < 30$ ms

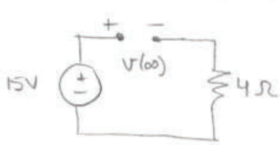


Step response :

$$v(t) = v(\infty) + [v(0) - v(\infty)]\, e^{-t/\tau}$$

where $v(0) = 0$ V

We calulate $v(\infty)$ (C is open circuit in steady state) :

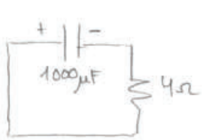


$v(\infty) = 15$ V

We calculate τ : $\tau = RC = 4\Omega \times 1000\,\mu F = 4$ms

Step response is then: $v_c(t) = 15 - 15 e^{-t/0.004}$ V for $0 < t < 30$ ms

- $30\text{ ms} < t < 60$ms



Natural response

$v(t) = v(t_0)\, e^{-(t-t_0)/RC}$, where $t_0 = 30$ms

We calculate $v(30\text{ms})$ from equation above : $v(30\text{ms}) = 15 - 15 e^{-0.03/0.004} = 14.991 \simeq 15$V

We calculate τ : $\tau = RC = 4$ms

Natural response is then: $v_c(t) = 15 e^{-(t-0.03)/0.004}$ for $30\text{ms} < t < 40$ms

QUESTION 3 [25 marks]

a. (12 marks) The two channels of a stereo amplifier are fed the same mono audio signal, v_s. A loudspeaker is connected between the two amplifier outputs, bridging the output terminals, as shown in Figure 5. This circuit, known as bridge amplifier, is used to increase the amount of power available in the speaker. Assuming ideal op amps:

- **i. (4 marks)** Derive an expression for the top op-amp output voltage v_1 in terms of the input voltage v_s.
- **ii. (4 marks)** Derive an expression for the bottom op-amp output voltage v_2 in terms of the input voltage v_s.
- **iii. (4 marks)** Derive an expression for the voltage gain v_0/v_s.

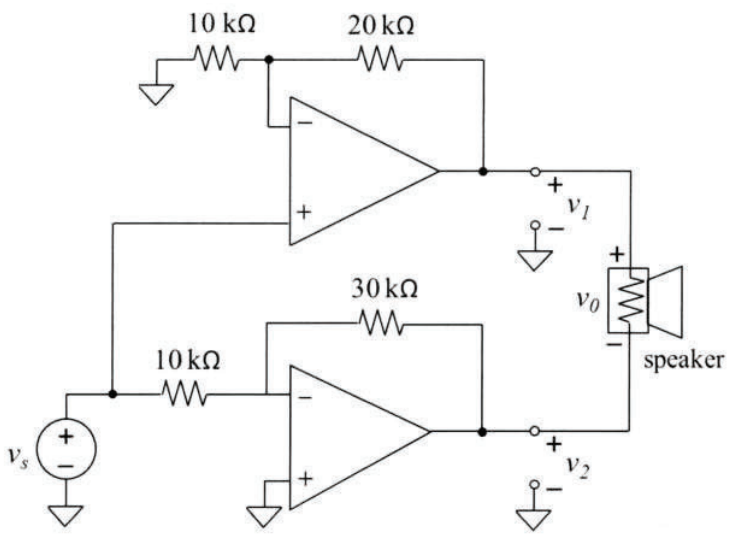
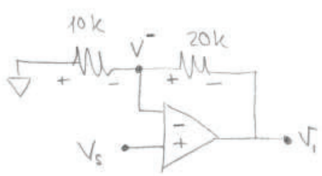


Figure 5

i)



Non-inverting amplifier

$V_{out} = \left(1 + R_f/R_1\right) V_{in}$

For our circuit:

$\boxed{V_1 = \left(1 + \frac{20K}{10K}\right) V_s = 3V_s}$

Alternatively, using nodal analysis:

KCL @ V^- : $\frac{0 - V^-}{10K} = \frac{V^- - V_1}{20K}$

Also, due to negative feedback:

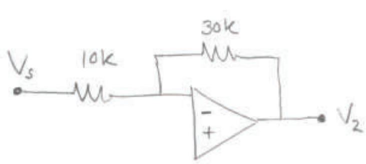
$V^+ = V^- = V_s$

Substituting : $\frac{-V_s}{10} = \frac{V_s}{20} - \frac{V_1}{20}$

$-2V_s = V_s - V_1$

$V_1 = 3V_s$

ii)



Inverting amplifier : $V_{out} = -R_f/R_1 \, V_{in}$

For our circuit: $\boxed{V_2 = -\frac{30K}{10K} V_s = -3V_s}$

iii) Use KVL to calculate V_o :

$-V_1 + V_o + V_2 = 0 \implies V_o = V_1 - V_2 = 3V_s - (-3V_s) = 6\,V_s$

$\Downarrow$

$\boxed{V_s/V_o = 6}$

b. (13 marks) Calculate $\mathbf{V_0}$ in the circuit shown in Figure 6.

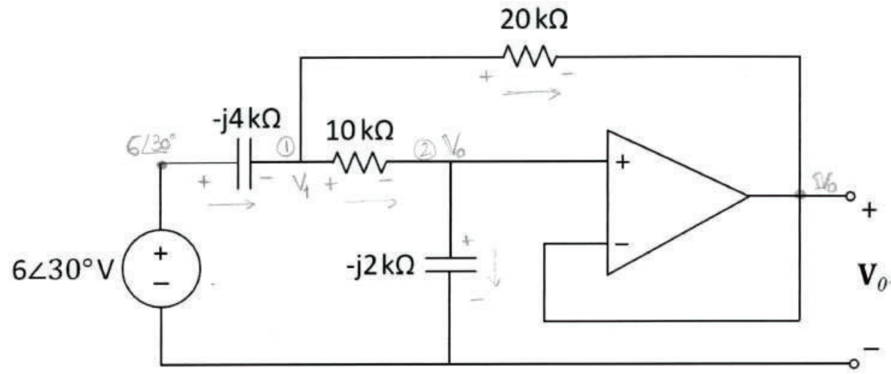


Figure 6

* KCL @ node 1 :

$$\frac{6\angle 30^\circ - V_1}{-j4k} = \frac{V_1 - V_0}{10k} + \frac{V_1 - V_0}{20k}$$

$$\frac{6\angle 30^\circ - V_1}{-j4} = \overbrace{\left(\frac{1}{10}+\frac{1}{20}\right)}^{0.15}(V_1 - V_0) \; ; \quad 6\angle 30^\circ = 0.15(-j4)(V_1 - V_0) + V_1 \; ; \quad 6\angle 30^\circ = (1 - j0.6)V_1 + j0.6V_0 \quad (1)$$

* KCL @ node 2 or voltage division to calculate V_0 :

$$V_0 = \frac{-j2k}{(10-j2)k} V_1 \implies V_1 = \frac{10-j2}{-j2} V_0 = \frac{10.2\angle -11.31^\circ}{2\angle -90^\circ} V_0 = 5.1\angle 78.69^\circ \times V_0 \quad (2)$$

voltage divider

If KCL is used: $\frac{V_1 - V_0}{10k} = \frac{V_0}{-j2k} \to V_1 = j\frac{10}{2}V_0 + V_0 = (1+j5)V_0 = 5.1\angle 78.69^\circ \times V_0$

Substitute (2) in (1) :

$$6\angle 30^\circ = (1-j0.6)\cdot 5.1\angle 78.69^\circ V_0 + j0.6V_0$$

$$6\angle 30^\circ = (1.17\angle -30.96^\circ \times 5.1\angle 78.69^\circ + j0.6)V_0$$

$$6\angle 30^\circ = (5.967\angle 47.73^\circ + j0.6)V_0 \; ; \quad 6\angle 30^\circ = (4.01 + j4.42 + j0.6)V_0$$

$$\boxed{V_0} = \frac{6\angle 30}{4.01 + j5.02} = \frac{6\angle 30^\circ}{6.42\angle 51.38^\circ} = \boxed{0.934\angle -21.38 \text{ V}}$$

QUESTION 4 [25 marks]

a. **(13 marks)** The circuit shown in Figure 7 corresponds to a "woofer" speaker which reproduces only the lower frequencies of music. The circuit diagram for the system is shown below.

   i. **(3 marks)** Express the voltage at the amplifier output v_a as a function of time and as a phasor.

   ii. **(6 marks)** Find the voltage v_{out} across the speaker as a phasor.

   iii. **(4 marks)** Find the average power consumed by the 1Ω internal resistance of the filter inductor.

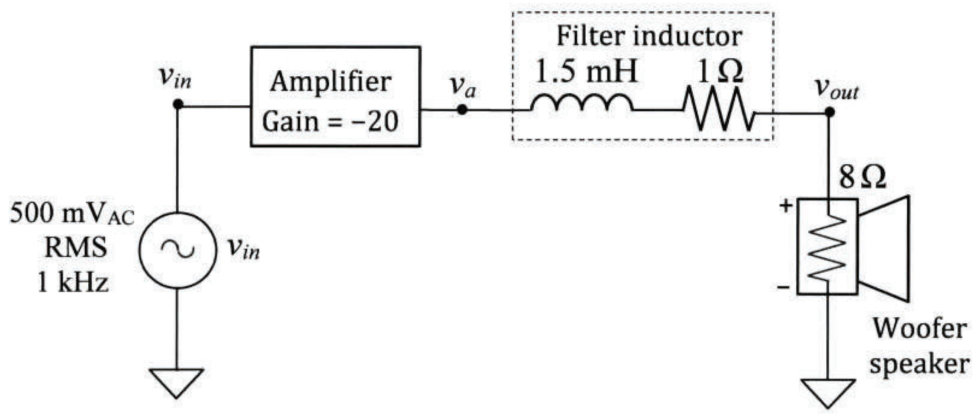


Figure 7

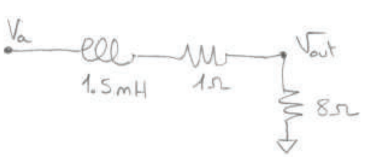
i)

$V_a = -20\,V_{in}$

$\boxed{V_a = -20 \times 0.5\sqrt{2}\cos(2000\pi t) = -14.14\cos(2000\pi t) = 14.14\cos(2000\pi t - 180^\circ)\ V}$

$\boxed{\mathbb{V}_a = 14.14\angle{-180^\circ}\ V}$

ii)



① Convert to phasor domain:

$Z_{1.5mH} = j\omega L = j\,2000\pi \times 1.5\times10^{-3} = j\,3\pi = j\,9.42\,\Omega$

$Z_{1\Omega} = 1\Omega$

$Z_{8\Omega} = 8\Omega$

② Use voltage division to find V_{out}:

$$\boxed{\mathbb{V}_{out} = \mathbb{V}_a \frac{8}{j9.42 + 1 + 8}} = 14.14\angle{-180^\circ} \times \frac{8}{9 + j9.42} = \frac{113.12\angle{-180^\circ}}{13.03\angle{43.31^\circ}} = 8.68\angle{-223.31} = \boxed{8.68\angle{136.69^\circ}\ V}$$

iii) $P_{avg} = \frac{1}{2}V_m I_m = \frac{1}{2}\frac{V_m^2}{R} = \frac{1}{2}I_m^2 \times R$ or $P_{avg} = V_{rms}I_{rms} = \frac{V_{rms}^2}{R} = I_{rms}^2 \times R$

$\mathbb{I} = \frac{\mathbb{V}_{out}}{8\Omega} = \frac{8.68\angle{136.69^\circ}}{8} = 1.085\angle{136.69^\circ}\ A \Rightarrow \boxed{P_{avg} = \frac{1}{2}\,1.085^2 \times 1 = 0.589\ W}$

b. (12 marks) In the circuit of Figure 8, determine the average power absorbed by the 40-Ω resistor.

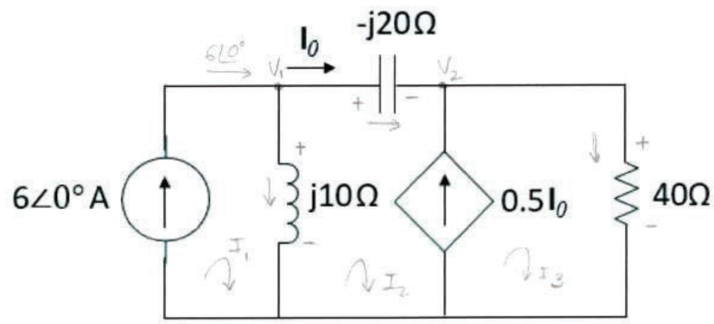


Figure 8

* Using nodal analysis:

* KCL @ node 1:

$6 = \frac{V_1}{j10} + \frac{V_1 - V_2}{-j20}$; $6 = \frac{V_1}{j10} + \frac{V_1}{-j20} - \frac{V_2}{-j20}$; $6 = (-j0.1 + j0.05)V_1 - j0.05V_2$

$6 = -j0.05V_1 - j0.05V_2$

$120 = -jV_1 - jV_2$

$V_1 = \frac{120 + jV_2}{-j} = 120j - V_2$ ①

* KCL @ node 2:

$I_o + 0.5I_o = \frac{V_2}{40}$; $1.5I_o = \frac{V_2}{40}$

Since $I_o = \frac{V_1 - V_2}{-j20}$ $\Rightarrow$ $\frac{1.5(V_1 - V_2)}{-j20} = \frac{V_2}{40}$

$1.5V_1 - 1.5V_2 = -j0.5V_2$; $1.5V_1 = (1.5 - j0.5)V_2$

$V_1 = (1 - j/3)V_2$ ②

Make ① and ② equal:

$120j - V_2 = (1 - j\frac{1}{3})V_2$

$120\angle 90° = (2 - j1/3)V_2 \Rightarrow V_2 = \frac{120\angle 90°}{2 - j1/3} = \frac{120\angle 90°}{2.03\angle -9.46°} = 59.11\angle 99.46°$

$V_{40\Omega} = V_2 = 59.11\angle 99.46°$

$\boxed{P_{40\Omega} = \frac{1}{2} V_m^2/R = \frac{1}{2} \times \frac{59.11^2}{40} = 43.68\ \text{W}}$

IGNORE THIS QUESTION

QUESTION 5 [10 marks]

a. **(2 marks)** Given the logic diagram in Figure 9, derive the logical expression for Z.

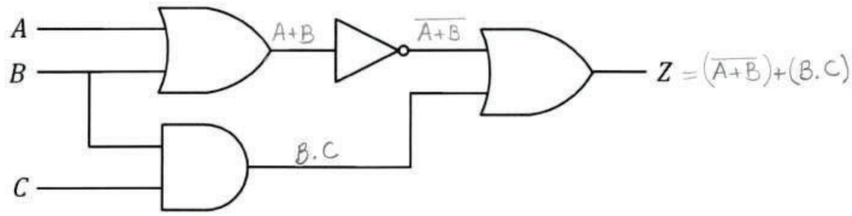


Figure 9

b. **(8 marks)** Consider the following identity:

$$\overline{(A+B).(\bar{A}+A.B)} = \bar{B}$$

i. **(5 marks)** Prove the identity using a truth table.

ii. **(3 marks)** Draw the logic diagram which represents the first term of the identity (i. e. $\overline{(A+B).(\bar{A}+A.B)}$).

END OF PAPER

a) $Z = (\overline{A+B})+(B.C)$ (as seen in Figure 9)

b) i)

A	B	A+B	AB	$\bar{A}$ + AB	(A+B).($\bar{A}$+AB)	$\overline{(A+B).(\bar{A}+AB)}$	$\bar{B}$
0	0	0	0	1	0	1	1
0	1	1	0	1	1	0	0
1	0	1	0	0	0	1	1
1	1	1	1	1	1	0	0

=

ii)